

MANOJ KUMAWAT

Senior Backend Engineer | Node.js · TypeScript · GraphQL · AWS

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SUMMARY

Senior Backend Engineer, 5+ years across product and security engineering — including 4+ years building e-commerce and subscription systems serving 1.6M+ users and 400K+ active subscriptions. Expertise in Node.js/TypeScript, GraphQL (Apollo) and REST API design, event-driven architecture, and AWS: led a full Shopify platform migration, cut cloud spend ~50%, and run a payout pipeline processing \$150K+/month. Cybersecurity background (OWASP Top 10, VAPT) — architect systems, code, and business logic secure-by-design: hardened authentication, scoped authorization, rate limiting, and query-complexity/abuse limits.

EXPERIENCE

Springworks

March 2022 – Present

Senior Backend Engineer — Burst Oral Care

Remote

- Led a full custom-platform to Shopify migration for a 400K+ subscription, 1.6M+ user store — implemented complex ambassador discount logic (40K+ codes) as Shopify Functions and cut over with no disruption to recurring billing.
- Sole DevOps owner for the platform — cut AWS spend 50% (Aurora/RDS tuning, ECS/Kubernetes right-sizing, CloudWatch retention/log-ingestion tuning, S3/Glacier lifecycle) and replaced a \$2K/month hosted BI tool (Periscope/Sisense) with self-hosted Redash on EC2, eliminating \$24K/year in licensing.
- Designed and built the ambassador payout system from scratch with idempotent Stripe transfers and automated reconciliation — a robust design that prevents duplicate transfers under retries and failures, across \$150K+/month.
- Optimized a high-load analytics pipeline by rewriting correlated subqueries into indexed JOINS on a 17M+ row activity table, cutting database load ~91% — diagnosed through slow-query analysis and composite index design.
- Built and operated REST APIs across subscriptions, payments, and orders (500K+ requests/day) on Aurora MySQL (12M+ row order tables), holding p95 latency under production peaks via connection pooling, indexing, and query tuning.
- Architected event-driven sync on Shopify's GraphQL Admin API and webhooks, mirroring customers, orders, products, and discounts into internal systems for near-real-time analytics and automation.
- Hardened the platform edge with Cloudflare — WAF rules, rate limiting, and Turnstile bot protection on abuse-prone endpoints, plus Workers and Pages for edge logic and static delivery; offloaded traffic from origin and reduced attack surface.
- Implemented Shopify Functions (cart validation, purchase restrictions, delivery customization, custom discounts), retiring paid third-party apps; built custom B2B/wholesale recurring billing Shopify could not natively support.
- Built custom MCP (Model Context Protocol) servers and connectors in TypeScript exposing internal systems (production database, Shopify, AWS) as structured, read-scoped tools — enabling safe AI-assisted operations and investigation over production data without direct credential exposure.

Truminds Software Systems

July 2020 – February 2022

Security Engineer — Penetration Testing & VAPT

Gurgaon, India

- Performed end-to-end VAPT (vulnerability assessment & penetration testing) on web applications, REST APIs, and network infrastructure — combining manual exploitation with Burp Suite, OWASP ZAP, sqlmap, Nessus, and Metasploit to uncover and exploit OWASP Top 10 flaws (SQL injection, broken access control, XSS, IDOR, auth bypass).
- Chained real-world attack paths — from injection and broken authentication to privilege escalation — and delivered detailed exploit walkthroughs, risk ratings (CVSS), and remediation guidance that engineering teams shipped to closure.
- Drove secure-by-design practices across teams — threat modeling, secure-coding reviews, and shift-left security in CI — eliminating recurring vulnerability classes and building the security instincts I now apply as a backend engineer.

PROJECTS

blue-node — backend reference implementation

2026

Node.js, TypeScript, Express 5, Apollo GraphQL, Postgres, MongoDB, Redis, AWS

github.com/imkumawat/blue-node

- Modular monolith in TypeScript (ESM) on Express 5 — per-module REST + GraphQL (Apollo) adapters over Postgres (Drizzle), MongoDB, and Redis, plus an MQTT 5 broker client as a resilient shared transport (auto-reconnect, graceful degradation); BullMQ + standalone SQS worker for background jobs.
- Per-request DataLoader batching to eliminate resolver N+1 (order-preserving batch queries behind a request-scoped context), with query-complexity limits on incoming operations.

- Directive-driven GraphQL security enforcement at the schema level — `@authenticated`, `@requireScope`, `@rateLimit`, and alias-abuse protection.
- Unified JWT auth core for REST + Apollo — access/refresh-token rotation & reuse detection, scoped permissions, and per-IP/account logout.
- Multi-instance real-time layer — WebSocket transport with a connection manager and Redis Pub/Sub fan-out (rooms) so events reach clients across horizontally-scaled instances; graceful shutdown drains in-flight requests and jobs on SIGTERM, with pino logging and CI (GitHub Actions + Azure DevOps) running typecheck, lint, and Vitest gates.

SKILLS

- **Languages:** TypeScript, JavaScript (Node.js, ES6+), Python
- **Backend & API:** GraphQL (Apollo), DataLoader, REST, WebSockets, Express 5, JWT
- **Data:** PostgreSQL, MySQL (Aurora), MongoDB, Redis
- **Messaging & Events:** AWS SQS/SNS/EventBridge, BullMQ, MQTT 5, event-driven architecture
- **Cloud & DevOps:** AWS (ECS/Fargate, EKS, Lambda, Aurora/RDS, ElastiCache, S3/Glacier, API Gateway, CloudWatch, Secrets Manager, IAM), Azure (Functions, Container Apps, Blob Storage, Front Door), Kubernetes, Docker, CI/CD (GitHub Actions, CodePipeline, Azure DevOps), Cloudflare (Workers, Pages, R2, WAF, Turnstile, DNS)
- **Observability:** Sentry, CloudWatch, OpenSearch, structured logging (pino)
- **Security:** OWASP Top 10, VAPT (Burp Suite, OWASP ZAP, Nessus, sqlmap, Metasploit), SonarCloud (SAST), secure coding, refresh-token reuse detection, rate limiting
- **AI & LLM Tooling:** Model Context Protocol (MCP) — custom servers & connectors, AI-assisted development, prompt engineering
- **Frontend:** Next.js, React

EDUCATION

Lovely Professional University
B.Tech in Computer Science & Engineering

Punjab, India
July 2016 – June 2020